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CIRCUITI E SISTEMI ELETTRONICI

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1 lezione

Books

□ Teoria

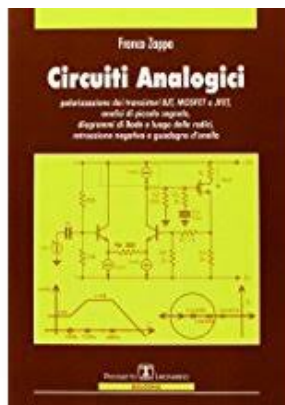
▣ Microelettronica

- Autori Richard c. Jaeger travis n.blalock
- Mc GrawHill



Books

- **Eserciziari**
 - ▣ **Circuiti analogici**
 - Autore F. Zappa
 - Progetto Leonardo
 - ▣ **Esercizi di progettazione elettronica**
 - Autori Andrea Bonfanti
 - Società editrice Esculapio



Program

- Introduction
 - ▣ Introduction analog design
 - ▣ Analog components
- Simple stages amplifier
 - ▣ Frequency analysis method
- Feedback loop
 - ▣ Analysis method
- Operational amplifier
 - ▣ Introduction
 - ▣ First stage
 - Differential Stage
 - Current mirrors
 - ▣ Second stage
 - Miller/Ahuja compensation
 - ▣ Output stages
- Non linear circuits
 - ▣ Bistable circuits and Oscillators

Introduzione Segnali

Segnale analogico e digitale

- *Segnale* è considerato qualsiasi valore misurabile di tensione, corrente o carica che descrive lo stato di un sistema o una grandezza fisica
 - ▣ *Segnale analogico* è definito in un continuo di valori di ampiezza e di tempo
 - ▣ *Segnale analogico campionato* è definito in un continuo di valori di ampiezza e in un insieme discreto di valori di tempo
 - ▣ *Segnale digitale* è definito in un insieme di valori discreti di tempo e di ampiezza

Introduzione Progetto e Analisi

Progetto e analisi

- Analisi è il processo estraee da uno schema circuitale le sue proprietà
 - ▣ Il risultato è unico
 - ▣ È realizzabile attraverso metodologie che possono essere automatizzate
- Progetto è il processo che parte dalle proprietà che sono definite come specifiche e termina con una soluzione circuitale
 - ▣ Il risultato non è unico è necessario introdurre delle cifre di merito per misurare le prestazioni
 - ▣ Il processo è automatizzabile solo parzialmente

Introduzione Organizzazione gerarchica del progetto

Ogni “prodotto” o “componente” è descritto da due e schemi:

- Schema funzionale: descrizione che contiene le informazioni mirate all' utilizzo del componente (data-sheet, manuale della tecnologia)
- Schema di realizzazione: descrizione che contiene le informazioni su come è costruito il componente basata su descrizioni funzionali dei componenti a livello gerarchico inferiore

Esempio

LM3100 SIMPLE SWITCHER® Synchronous 1MHz 1.5A Step-Down Voltage Regulator

General Description

The LM3100 Synchronously Rectified Buck Converter features all functions needed to implement a highly efficient, cost effective buck regulator capable of supplying 1.5A to loads with voltages as low as 0.8V. Dual 40V N-Channel synchronous MOSFET switches allow for low external component thus reducing complexity and minimizing board space. The LM3100 is designed to work exceptionally well with ceramic and other very low ESR output capacitors. The Constant ON-Time (COT) regulation scheme requires no loop compensation, results in fast load transient response, and simplifies circuit implementation. Through the use of a unique design the regulator does not rely on output capacitor ESR for stability, as do most other COT regulators. The operating frequency remains nearly constant with line and load variations due to the inverse relationship between the input voltage and the on-time. The operating frequency can be externally programmed up to 1MHz. Protection features include V_{CC} under-voltage lockout, thermal shutdown and gate drive under-voltage lockout. The part is available in a thermally enhanced eTSSOP-20 package

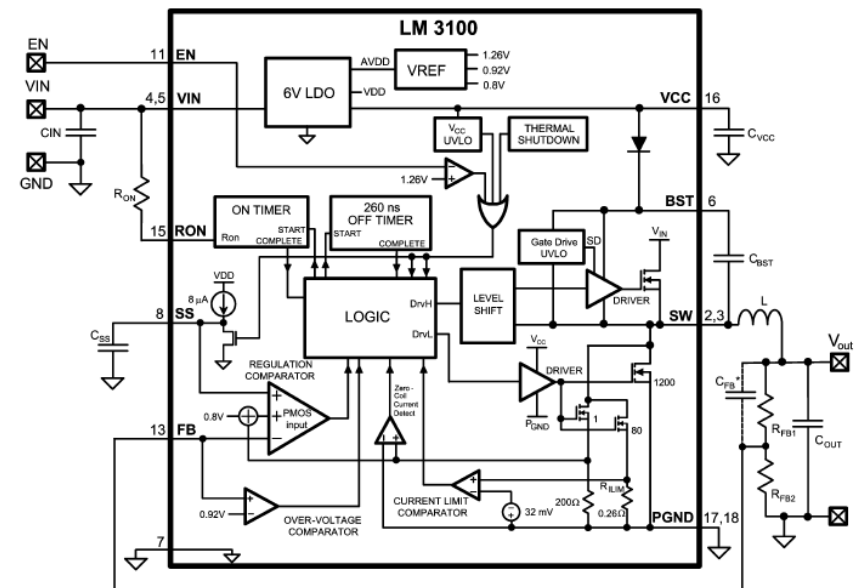
Features

- Input voltage range 4.5V - 36V
- 1.5A output current
- 0.8V, $\pm 1.5\%$ reference
- Integrated 40V, dual N-Channel buck synchronous switches
- Low component count and small solution size
- No loop compensation required
- Ultra-fast transient response
- Stable with ceramic and other low ESR capacitors
- Programmable switching frequency up to 1MHz
- Max. duty cycle limited during start-up
- Valley current limit
- Precision Internal Reference for adjustable output voltage down to 0.8V
- Thermal shutdown
- Thermally enhanced eTSSOP-20 package

Typical Applications

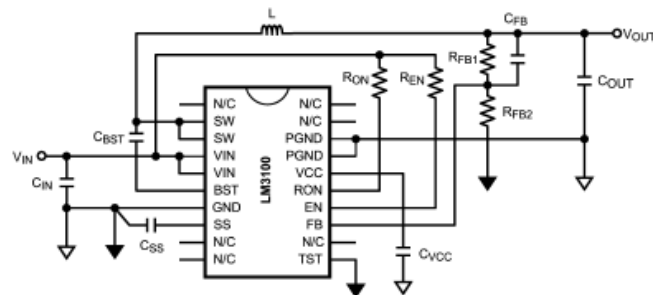
- 5VDC, 12VDC, 24VDC, 12VAC, and 24VAC systems
- Embedded Systems
- Industrial Controls
- Automotive Telematics and Body Electronics
- Point of Load Regulators
- Storage Systems
- Broadband Infrastructure
- Direct Conversion from 2/3/4 Cell Lithium Batteries Systems

Simplified Functional Block Diagram

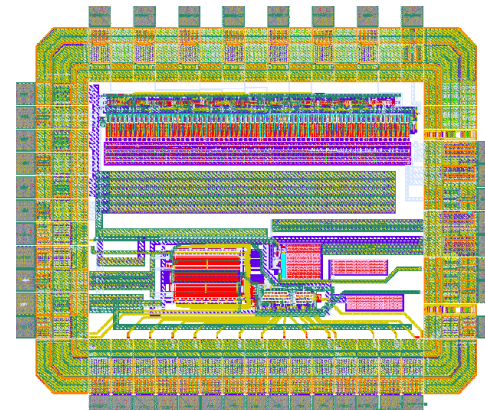


*optional
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Typical Application

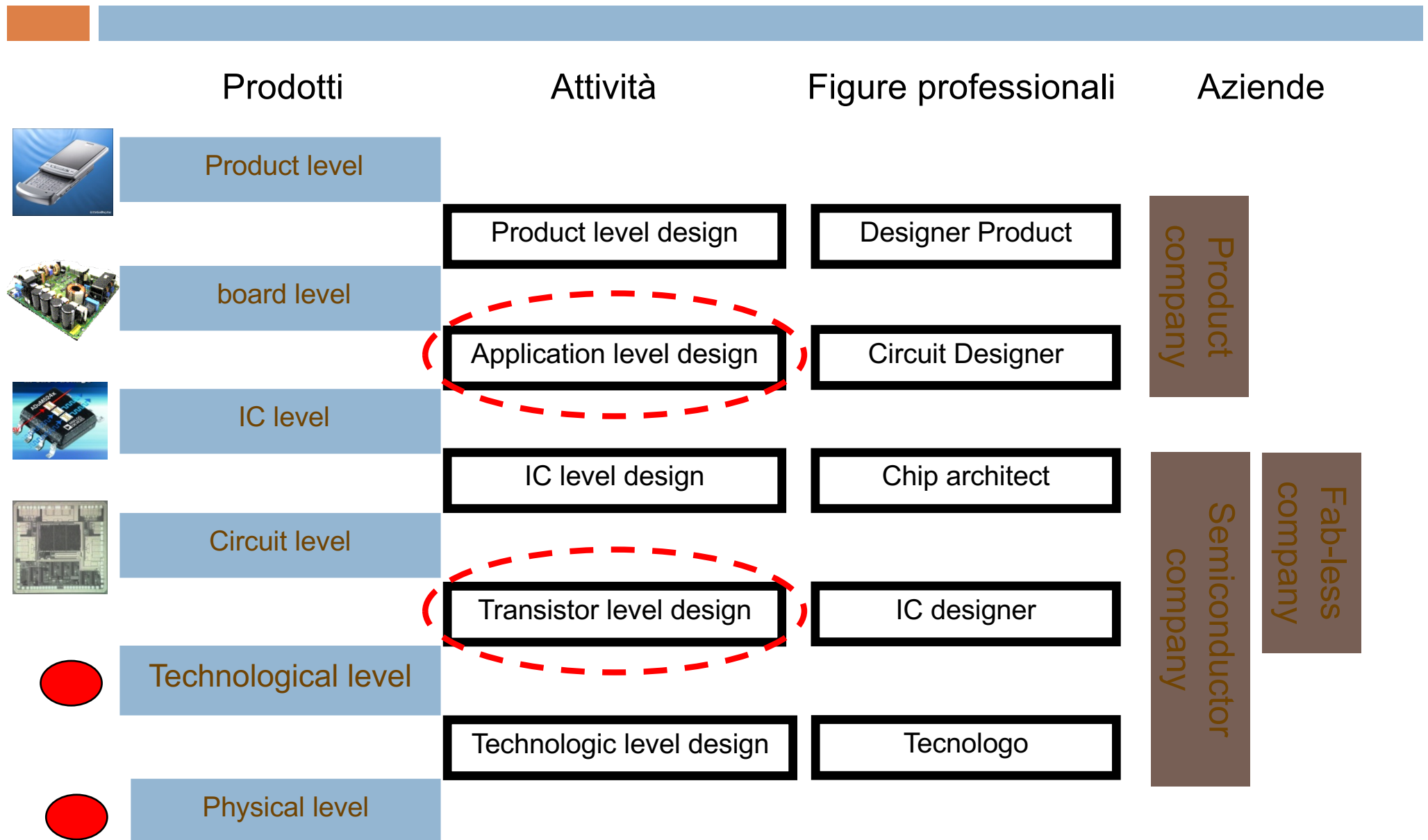


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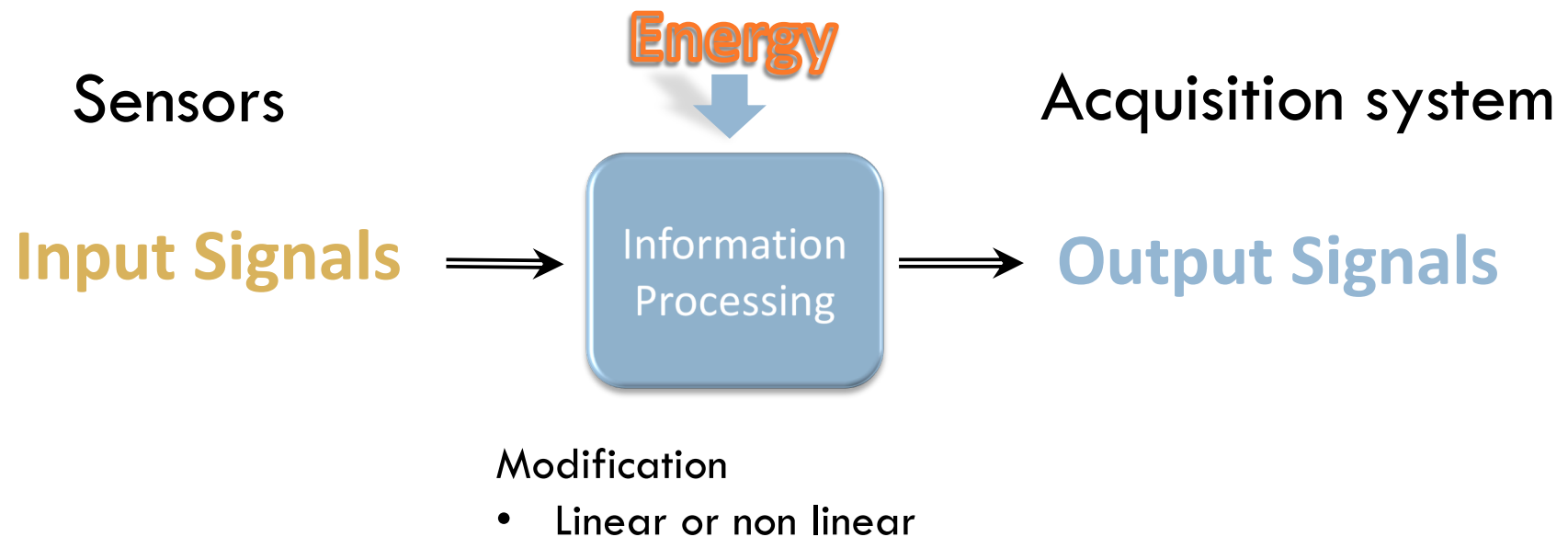
Introduzione

Organizzazione gerarchica del progetto



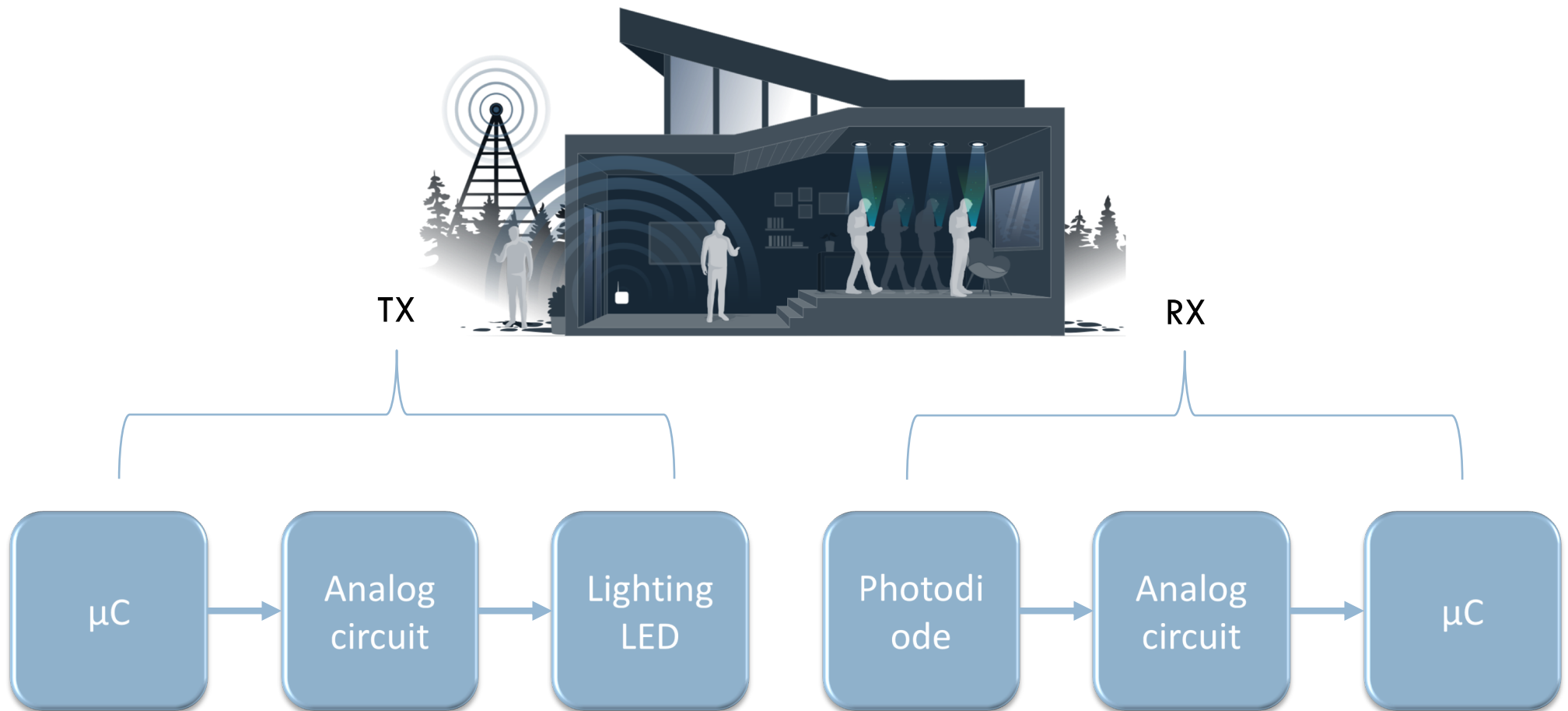
Example of design

- Discussing example related
 - ▣ Analog processing



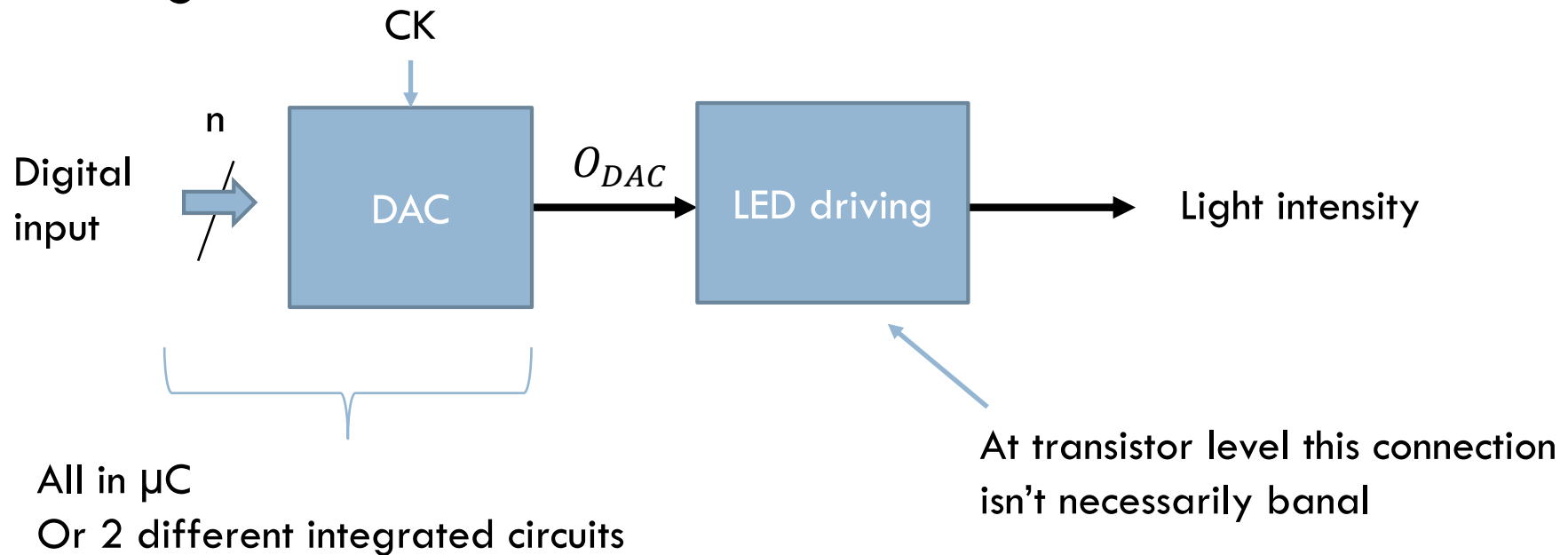
Example of design: LiFi

□ Specification



Design of TX

□ High level

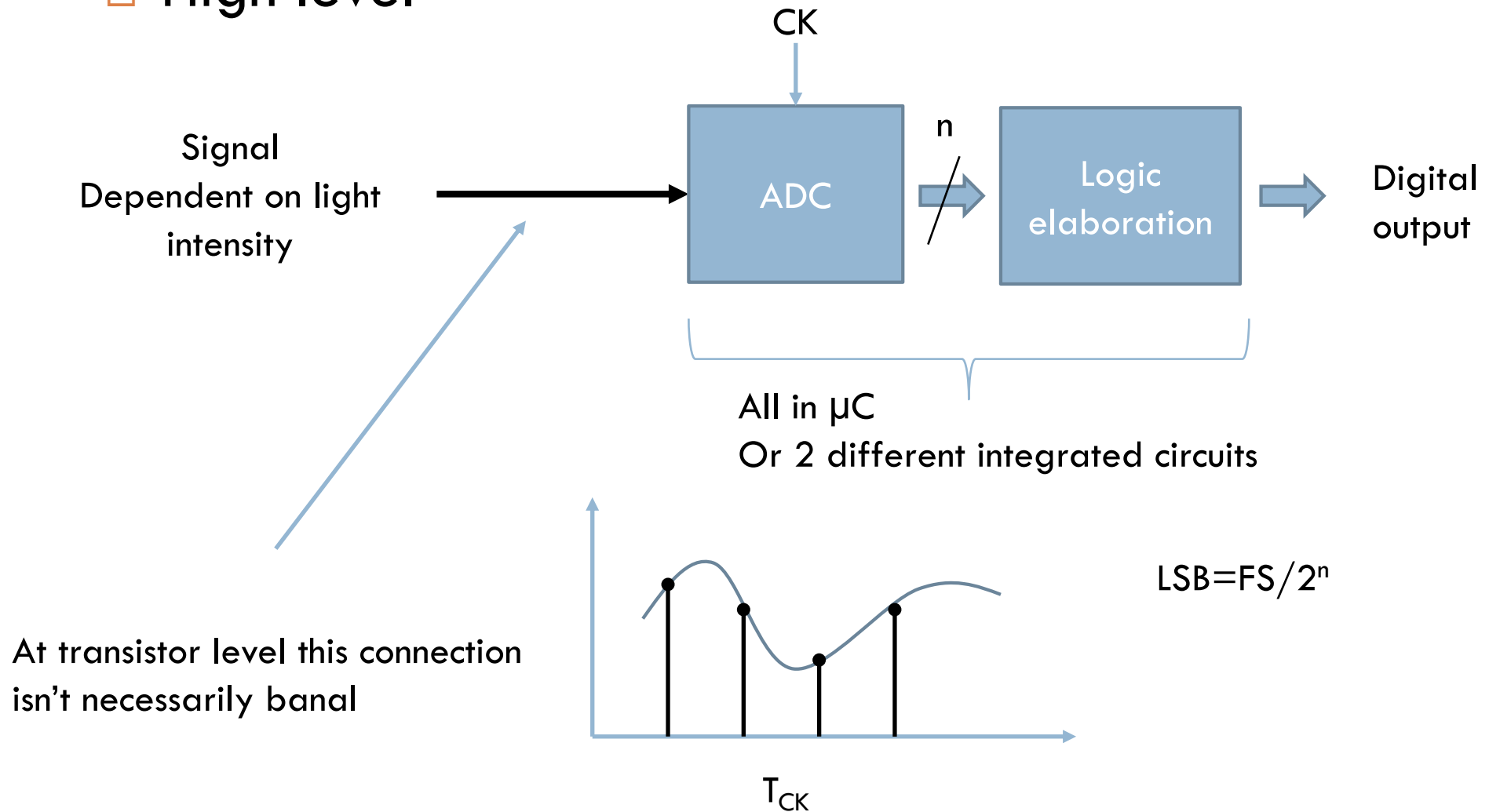


$$O_{DAC} = FS (b_1 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-n})$$

Dove $b_i \in \{0,1\}$

Design of RX

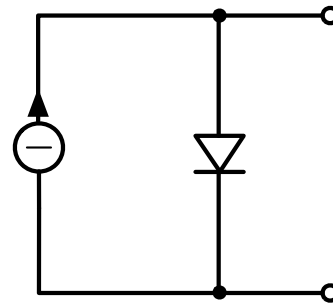
□ High level



Example of sensor: Photodiode

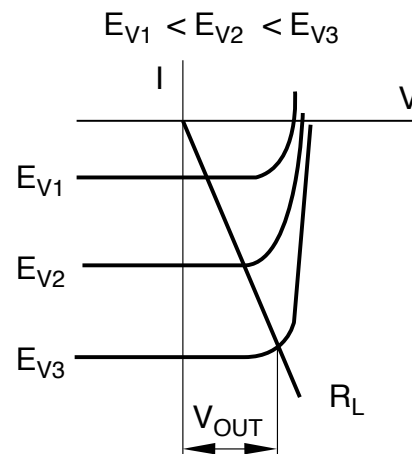
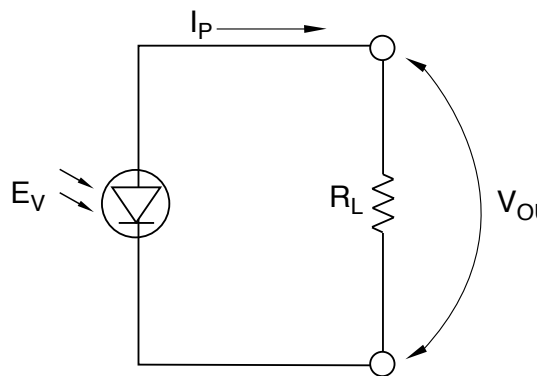
□ Sensor photodiode

Proportional to
The amount of
Incident light

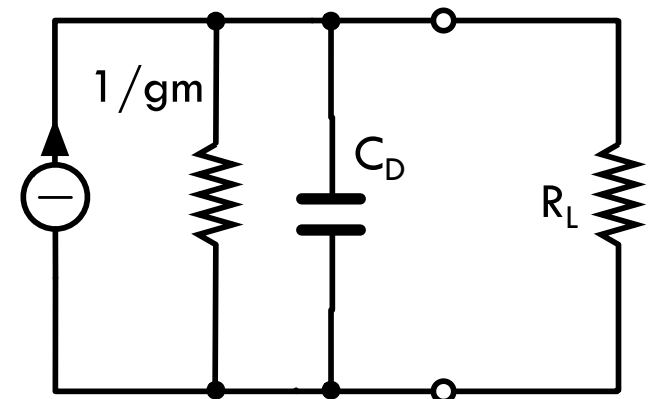


Large signal model

Large signal analysis

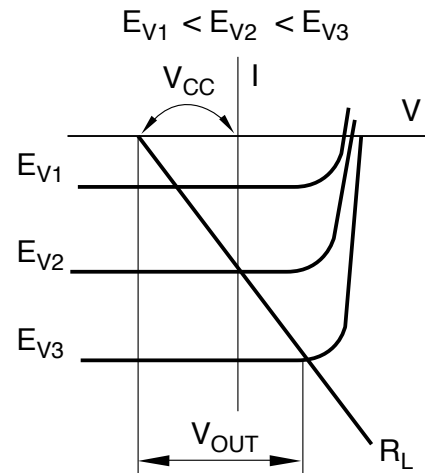
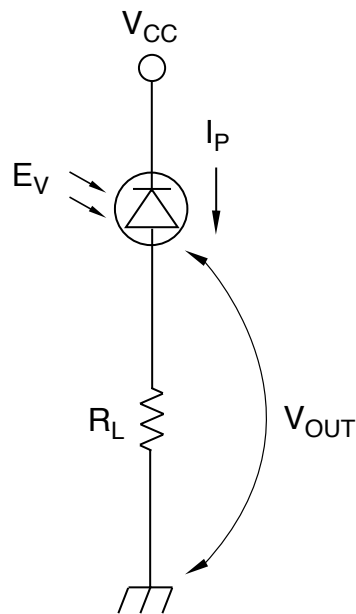


Small signal model

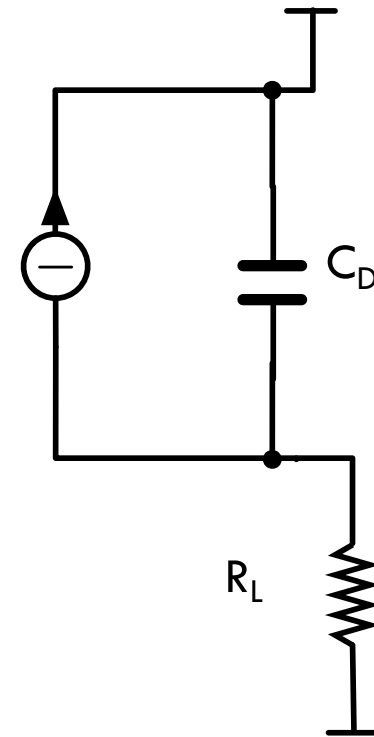


Example Photodiode

□ Polarized inverse

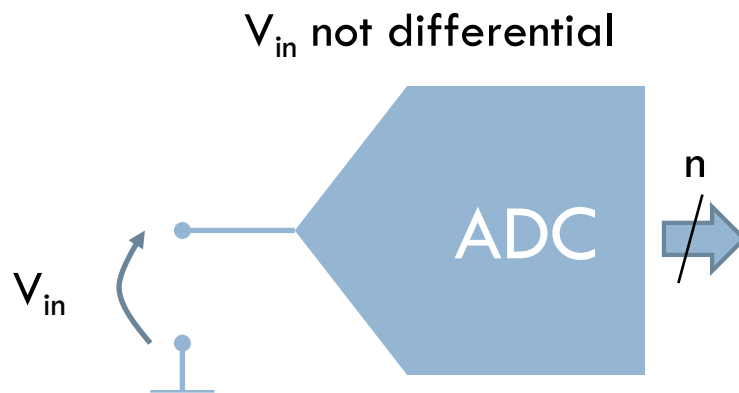


Small signal model

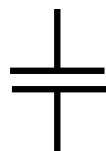


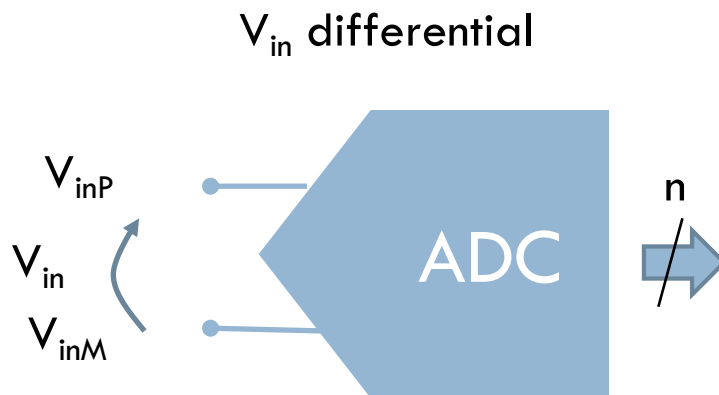
Basic ADC

□ ADC conversion two types



V_{in} input conversion range
From a $V_{in\ MIN}$ to a $V_{in\ MAX}$ with n bit

Z_{in}  Input impedance



Each input has a dynamic range
from a $V_{P,M\ MIN}$ to a $V_{P,M\ MAX}$

The difference $V_{in} = V_{in\ P} - V_{in\ M}$
is converted From a $V_{in\ MIN}$ to a $V_{in\ MAX}$
with n bit